Exercise 3b)

$$C = \frac{1}{N} \sum_{i=1}^{N} \left(y_i - \hat{y}_i \right)^2$$

$$= \frac{1}{N} \sum_{i=1}^{N} \left(y_i - a_i \right)^2$$

$$= \frac{1}{N} \sum_{i=1}^{N} \left(y_i^2 - 2 y_i a_i + a_i^2 \right)$$

$$\frac{\partial C}{\partial a_i} = \frac{1}{N} \sum_{i=1}^{N} \left(-2 y_i + 2 a_i \right)$$

$$= \frac{2}{N} \sum_{i=1}^{N} \left(a_i - y_i \right)$$

$$\frac{\partial C}{\partial a} = \frac{2}{N}\left[\vec{a} - \vec{y}\right]$$

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3 c)

$$a = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma'(z) = U'(z) \cdot \sigma'(V)$$

$$= \left(1 + e^{-z}\right)' \cdot \left(\frac{1}{V}\right)'$$

$$= -e^{-z} \cdot -\frac{1}{V^2}$$

$$= \frac{e^{-z}}{\left(1 + e^{-z}\right)^2}$$

$$= \frac{1}{(1+e^{-z})} \cdot \frac{e^{-z}}{(1+e^{-z})}$$

$$\sigma(z) = \frac{1}{(1+e^{-z})}$$

$$e^{-z} = \frac{1}{\sigma(z)} - 1$$

$$= \left[\frac{1}{(1+e^{-z})}\right]^2 \cdot e^{-z}$$

$$= \sigma(z)^2 \cdot \left(\frac{1}{\sigma(z)} - 1\right)$$

$$= \underline{\underline{\sigma(z)\left(1 - \sigma(z)\right)}}$$

3 e)

$$z = Wx + b$$

$$= \sum_i \sum_j W_{ij} x_j + b_i$$

$$z_i = \sum_j W_{ij} x_j + b_i$$

$$\frac{\partial z_i}{\partial b_k} = \delta_{ik}$$

$$\underline{\underline{\frac{\partial z}{\partial b} = I}}$$

$$z = Wx + b$$

$$z = \sum_{ij} W_{ij} x_j + b_i$$

$$z_i = \sum_j W_{ij} x_j + b_i$$

$$\frac{\partial z_i}{\partial W_{kl}} = \sum_j \delta_{ik} \delta_{jl} x_l$$

$$= \delta_{ik} x_l$$

$$\frac{\partial z}{\partial W} = I \otimes \vec{x}$$

$$\frac{\partial C}{\partial b} = \frac{\partial C}{\partial a} \frac{\partial a}{\partial z} \frac{\partial z}{\partial b}$$

$$= \frac{\partial C}{\partial a} \frac{\partial a}{\partial z} \cdot I$$

$$= \frac{\partial c}{\partial a} \frac{\partial a}{\partial z}$$

$$\frac{\partial c}{\partial w} = \frac{\partial c}{\partial a} \frac{\partial a}{\partial z} \frac{\partial z}{\partial w}$$

$$= \frac{\partial c}{\partial a} \frac{\partial a}{\partial z} \cdot (I \otimes \hat{X})$$

$$= \frac{\partial c}{\partial a} \frac{\partial a}{\partial z} \cdot X^T$$

$\frac{\partial z}{\partial w}$ → 2D identity Matrix

$\frac{\partial b}{}$ of size output x output

$\frac{\partial z}{\partial W}$ → 3D Matrix of size output x input x output.

Exercise 4b)

$$z^1 = W^1 a^0 + b^1 \qquad (a^0 = x)$$

$$a^1 = f(z^1)$$

$$z^2 = W^2 a^1 + b^2$$

$$z^2_i = \sum_j W_{ij} a^1_j + b^2_i$$

$$\frac{\partial z^2_i}{\partial a^1_k} = W_{ik}$$

$$\frac{\partial z^2}{\partial a^1} = W^2$$

$$\frac{\partial z^{(l)}}{\partial a^{(l-1)}} = W^{(l)}$$

4c) $$\frac{\partial C}{\partial a^1} = \frac{\partial C}{\partial z^2} \frac{\partial z^2}{\partial a^1}$$

$$\frac{\partial C}{\partial a^1_j} = \sum_i \frac{\partial C}{\partial z^2_i} \frac{\partial z^2_i}{\partial a^1_j}$$

$$\frac{\partial C}{\partial a^1_j} = \sum_i \frac{\partial C}{\partial z^2_i} W_{ij}$$

$$\left[\frac{\partial C}{\partial} \right] \rightarrow \sum \frac{\partial C}{\partial z^2_i} W_{i1}$$

$$\begin{bmatrix} \frac{\partial C}{\partial a_1^1} \\ \frac{\partial C}{\partial a_2^1} \\ \frac{\partial C}{\partial a_3^1} \end{bmatrix} \rightarrow \begin{matrix} \sum_i \frac{\partial C}{\partial z_i^2} W_{i1} \\ \sum_i \frac{\partial C}{\partial z_i^2} W_{i2} \\ \sum_i \frac{\partial C}{\partial z_i^2} W_{i3} \end{matrix}$$

$$\frac{\partial C}{\partial a^1} = W^T \frac{\partial C}{\partial z^2}$$